Data Mining Approach to Clinical Pattern Discovery & Mortality Risk Modeling

Course: DS-600 Data Mining
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Abstract

This project applies a data mining workflow to multimodal clinical data to (1) discover clinically meaningful patient phenotypes and (2) develop interpretable mortality risk models. Using the Symile-MIMIC dataset derived from MIMIC-IV, MIMIC-IV-ECG, and MIMIC-CXR-JPG, we constructed a clean analytic feature set consisting of 47 commonly ordered laboratory variables, 8 radiographic labels with adequate coverage, and 6 demographic/admission variables. Exploratory analysis identified strong physiologic separation between survivors and non-survivors (e.g., lactate, creatinine), a right-skewed length-of-stay distribution, and admission-type effects consistent with severity differences. Unsupervised clustering with PCA + KMeans revealed population-level clinical phenotypes with monotonic gradients in mortality and length of stay, including a high-acuity phenotype characterized by metabolic acidosis and renal dysfunction. Supervised models (logistic regression and random forest) achieved moderate but clinically meaningful discrimination (AUC ~0.77–0.78) and converged on consistent predictors aligned with critical illness physiology (shock/hypoperfusion, inflammation, renal dysfunction, coagulopathy, and frailty). Subgroup evaluation by gender and race showed broadly similar discrimination with modest variation. Ethical considerations related to fairness, privacy, and clinical deployment are discussed.

1. Introduction

Healthcare data mining aims to extract actionable insights from high-dimensional clinical data while maintaining interpretability, safety, and fairness. Mortality prediction is a clinically important task because early risk stratification can support triage, escalation of care, and targeted monitoring. However, predictive performance alone is insufficient in healthcare: responsible work must also emphasize **pattern discovery**, subgroup analysis, interpretability, and societal impacts.

Project goals:

1. Perform exploratory clinical pattern discovery (mortality and LOS distributions; key feature-outcome relationships).
2. Identify data-driven patient phenotypes using unsupervised learning and interpret them clinically.
3. Build mortality risk models and interpret drivers of risk (logistic regression + random forest).

4. Evaluate subgroup model discrimination (fairness-oriented analysis) and discuss ethical/privacy implications.

While machine learning is used for predictive modeling, this project emphasizes data mining through exploratory analysis, feature interpretation, subgroup comparison, and evaluation of societal impacts such as fairness, privacy, and clinical decision-making.

2. Dataset Description

The Symile-MIMIC dataset integrates admissions from MIMIC-IV, MIMIC-CXR-JPG, and MIMIC-IV-ECG. Each record includes:

- Blood laboratory measurements obtained within 24 hours of admission
- Chest X-ray labels derived using CheXpert methodology
- Demographic and admission information
- In-hospital mortality indicator (hospital_expire_flag)

Only admissions containing all three modalities (labs, ECG, and CXR) were retained.

After preprocessing, the analytic dataset contained **11,027 patient admissions** with:

- **47 laboratory variables**
- **8 radiographic labels** (Atelectasis, Cardiomegaly, Edema, Lung Opacity, Pleural Effusion, Pneumonia, Pneumothorax, Support Devices)
- **6 demographic/admission variables** (age, gender, race, admission type, admission location, discharge location)

The outcome of interest was in-hospital mortality. Length of stay (LOS) was computed as the difference between admission and discharge timestamps.

The observed mortality rate was approximately **9.2%**, consistent with ICU-adjacent cohorts.

3. Data Preprocessing

Several preprocessing steps were applied to prepare the data for analysis:

1. Identifiers, timestamps, file paths, and imaging metadata were removed as they are not analytically meaningful for modeling.
2. Variables with more than 80% missingness were excluded to reduce noise and dimensionality.
3. Remaining laboratory features were median-imputed.
4. Continuous variables were standardized prior to clustering and logistic regression.
5. Length of stay was derived from admission and discharge times.

Following preprocessing, the final modeling matrix contained 56 clinical features.

4. Exploratory Data Analysis

4.1 Outcome Overview

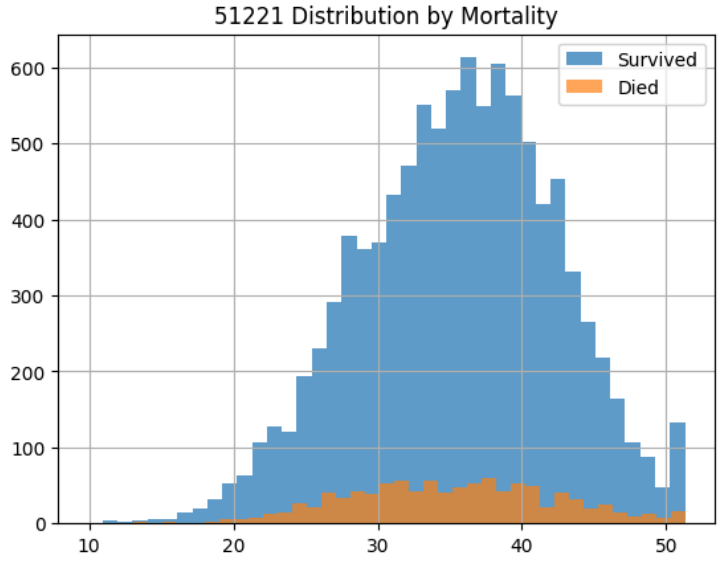
In-hospital mortality was 9.2%. Length of stay exhibited substantial right skew, with a median of approximately 6.8 days and a small subset of patients experiencing prolonged hospitalizations exceeding 30 days. LOS exhibits classic hospital behavior: most discharges occur within about a week, with a long tail of complex cases.

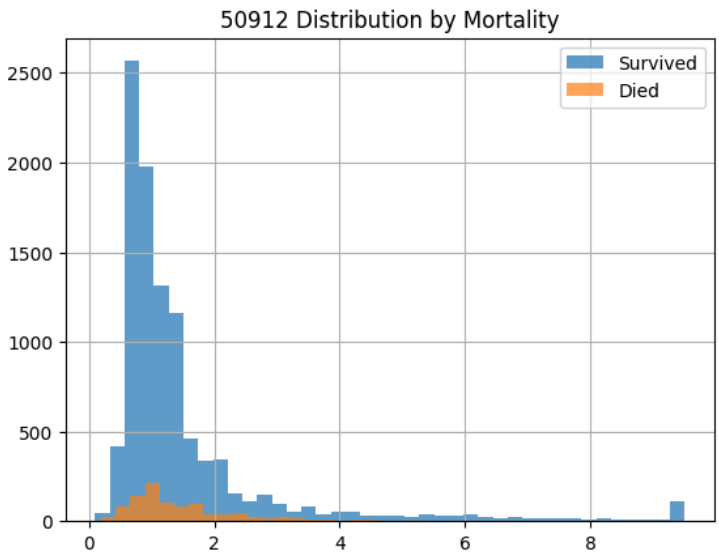
4.2 Laboratory Patterns and Mortality

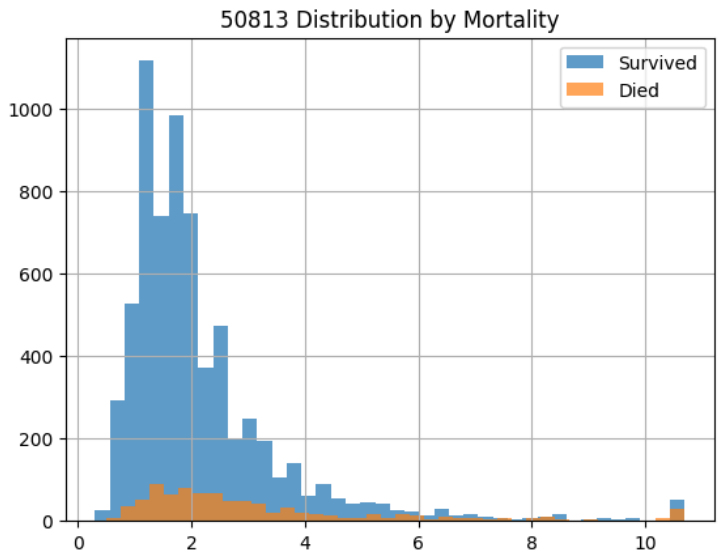
Several laboratory markers demonstrated clear separation between survivors and non-survivors:

- **Creatinine (50912):** Higher values among deceased patients, indicating renal dysfunction.
- **Lactate (50813):** Strong right shift in deceased patients, consistent with shock and tissue hypoperfusion.
- **Hematocrit (51221):** Lower central tendency in deceased patients, suggesting anemia or hemodilution.

These findings validate the clinical relevance of the dataset.







4.3 Demographic and Admission Characteristics

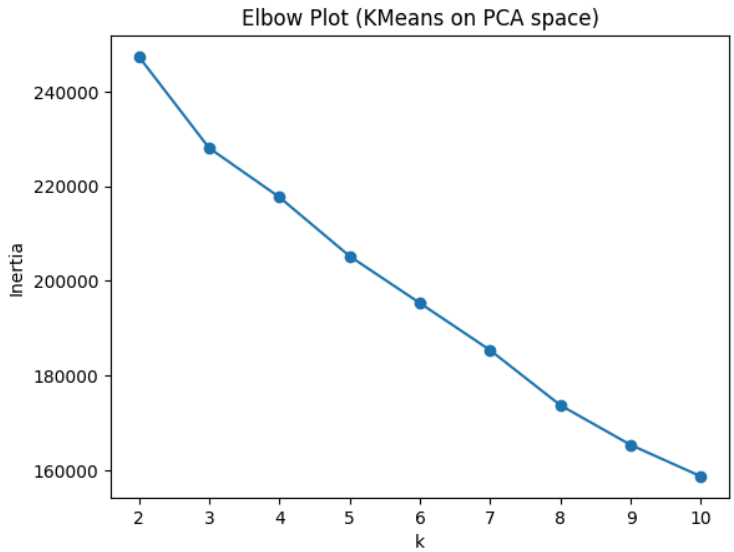
Mortality rates were similar across genders (male ≈ 9.5%, female ≈ 8.9%). Emergency admissions exhibited substantially higher mortality (~11%) compared to elective admissions (~1%), reflecting differences in acuity.

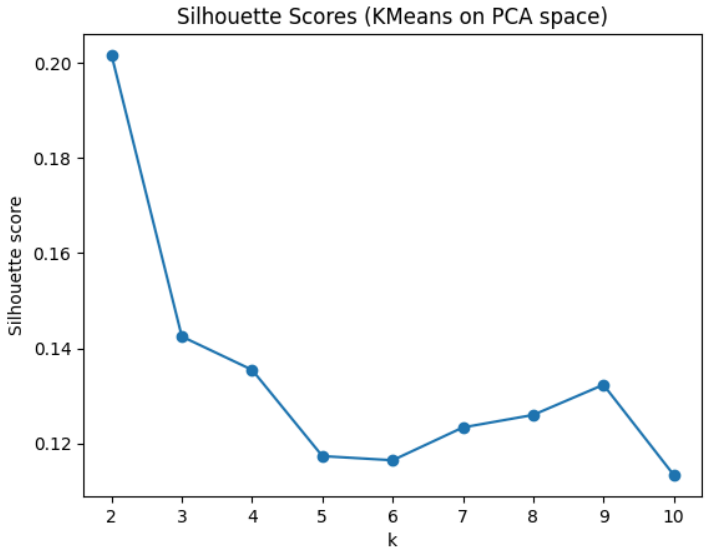
Categories such as “Unknown” or “Unable to obtain” race showed elevated mortality, likely reflecting documentation limitations in critically ill patients rather than intrinsic demographic effects.

5. Clinical Pattern Discovery (Unsupervised Learning)

To identify latent patient phenotypes, clustering was performed using age, laboratory values, and radiographic labels.

Because high-dimensional clinical data can lead to unstable clustering, Principal Component Analysis (PCA) was applied prior to KMeans. The first 10 components explained approximately 45% of total variance.





	k	inertia	silhouette
0	2	247365.184729	0.201596
1	3	228061.071305	0.142469
2	4	217672.460383	0.135461
3	5	205202.461022	0.117343
4	6	195290.202337	0.116467
5	7	185324.473579	0.123367
6	8	173695.146849	0.125991
7	9	165317.597010	0.132332
8	10	158660.873672	0.113385

Elbow and silhouette analyses supported a four-cluster solution.

5.1 Cluster Characteristics

Four patient clusters were identified:

- **Cluster 1 (Low Risk):** Largest group (~50%), lowest mortality (~5.7%), shortest LOS.
- **Cluster 2 (Moderate Risk):** Older patients with moderate mortality (~9.6%).
- **Cluster 0 (High Risk):** Elevated mortality (~14.6%) with longer LOS.
- **Cluster 3 (Extreme Outliers):** Small group (~5%) with very high mortality (~27.9%).

Cluster 3 was excluded from detailed interpretation due to small sample size.

5.2 Physiologic Profiles

The high-risk cluster demonstrated:

- Elevated creatinine and blood urea nitrogen (renal dysfunction)
- Increased lactate and anion gap with decreased pH (metabolic acidosis)
- Elevated white blood cell counts (systemic inflammation)
- Prolonged coagulation markers

These findings represent a classic multi-organ dysfunction phenotype commonly observed in critically ill patients.

This unsupervised discovery of clinically coherent subgroups demonstrates the value of data mining beyond simple prediction.

6. Mortality Prediction Modeling

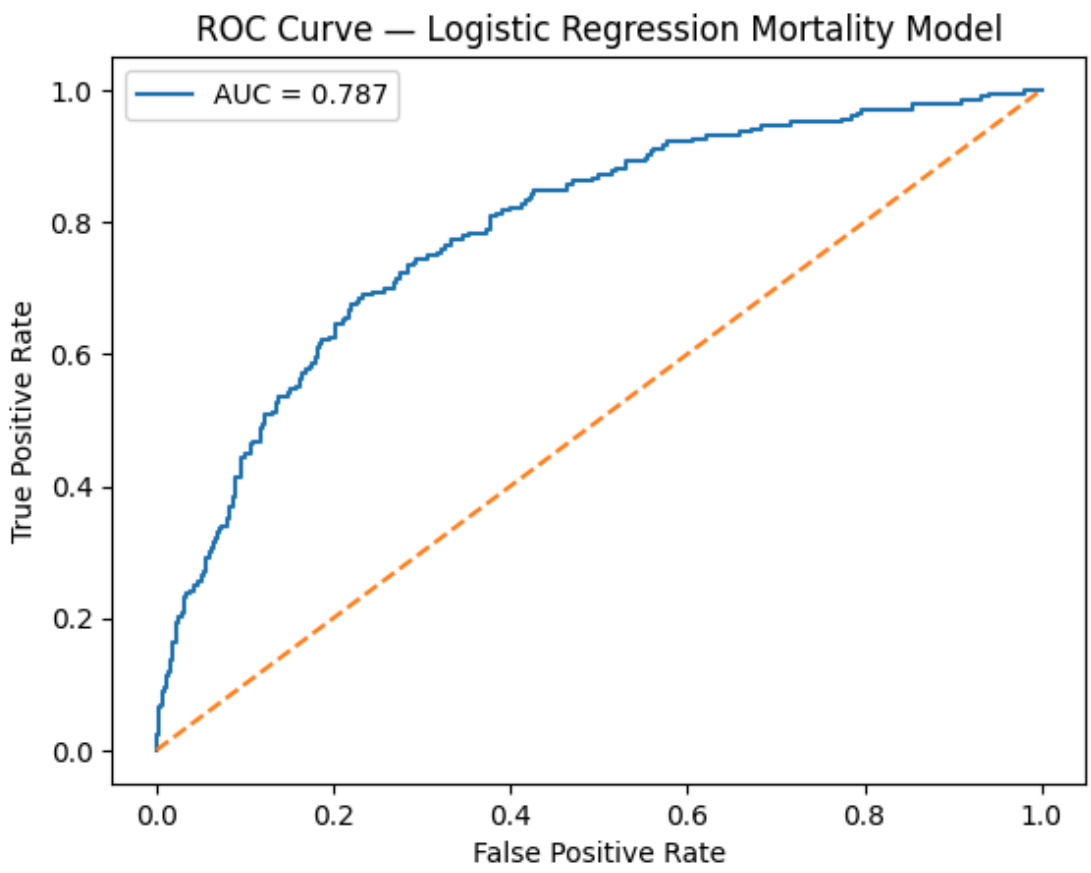
Two supervised models were developed using early laboratory values and age:

1. Logistic Regression (with class balancing)
2. Random Forest

6.1 Model Performance

- Logistic Regression AUC: **0.787**
- Random Forest AUC: **0.805**

Both models achieved moderate but clinically meaningful discrimination. The small performance gap indicates that much of the signal is captured by linear relationships.

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Logistic Regression Test AUC: 0.7871198128915001
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6.2 Logistic Regression Interpretation

Top positive predictors of mortality (per standard deviation increase):

- Prothrombin Time (PT)
- Red Blood Cell indices
- Hemoglobin

- Age
- Lactate
- RDW
- Anion Gap
- Blood Urea Nitrogen
- pH

These features reflect coagulopathy, metabolic acidosis, renal dysfunction, inflammation, and frailty.

	feature	coef	name
0	51274	1.191555	Prothrombin Time (PT)
1	51279	1.064140	51279
2	51222	0.602614	51222
3	51248	0.566502	51248
4	age	0.415172	age
5	50813	0.361224	Lactate
6	51277	0.272422	RDW
7	50818	0.260842	50818
8	50861	0.240020	50861
9	50821	0.212605	50821
10	50868	0.204419	Anion Gap
11	51006	0.147498	Blood Urea Nitrogen (BUN)
12	50863	0.121789	50863
13	52073	0.112499	52073
14	50820	0.109536	pH

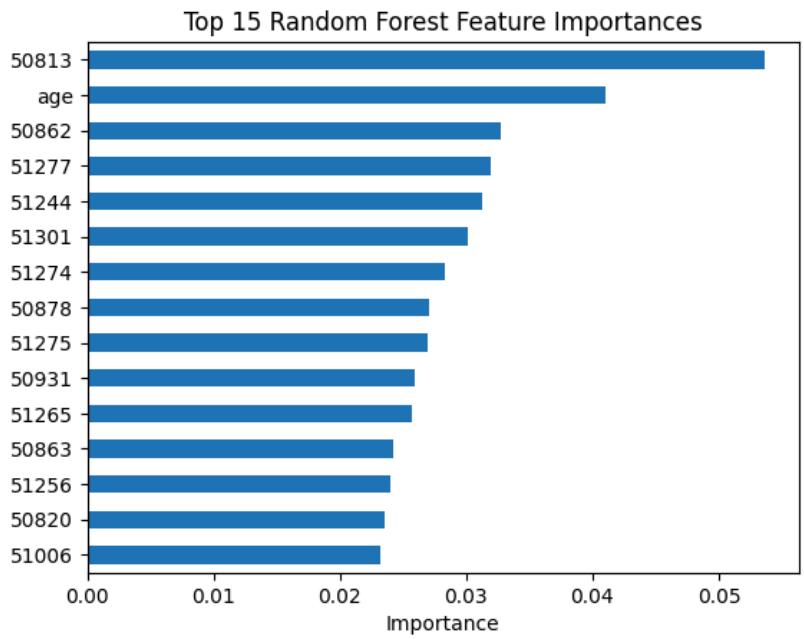
6.3 Random Forest Feature Importance

The Random Forest model identified:

- Lactate
- Age
- Albumin
- RDW

- Lymphocytes
- White Blood Cell Count
- PT/PTT
- AST
- Glucose
- BUN

as the most influential predictors.



	feature	importance	name
0	50813	0.053652	Lactate
1	age	0.041041	age
2	50862	0.032705	Albumin
3	51277	0.031925	RDW
4	51244	0.031306	Lymphocytes
5	51301	0.030164	White Blood Cell Count (WBC)
6	51274	0.028256	Prothrombin Time (PT)
7	50878	0.027118	AST
8	51275	0.026942	Partial Thromboplastin Time (PTT)
9	50931	0.025873	Glucose
10	51265	0.025757	51265
11	50863	0.024245	50863
12	51256	0.024050	Neutrophils
13	50820	0.023535	pH
14	51006	0.023226	Blood Urea Nitrogen (BUN)

6.4 Cross-Model Consistency

Both models consistently highlighted:

- Shock markers (lactate, pH)
- Renal dysfunction (BUN)
- Coagulation abnormalities (PT/PTT)
- Inflammatory response (WBC, neutrophils, lymphocytes)
- Frailty indicators (age, albumin, RDW)

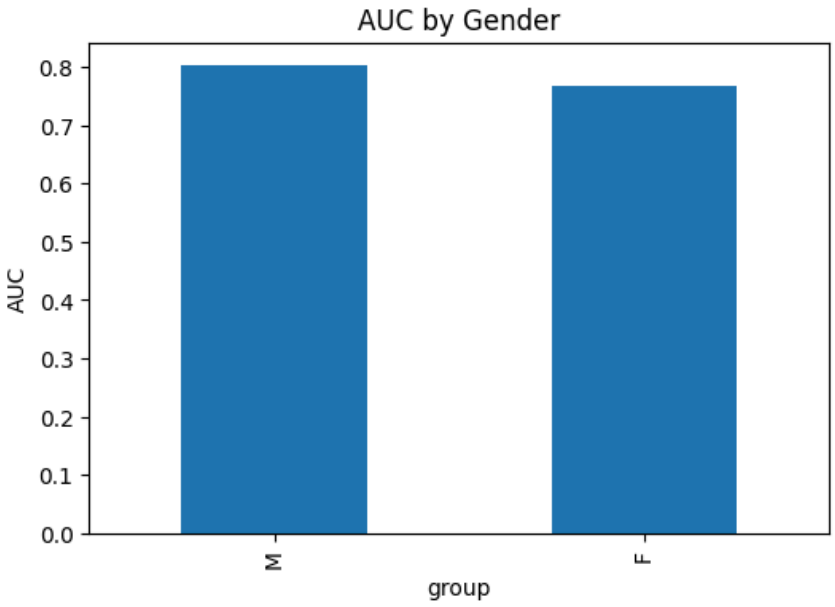
This agreement across unsupervised clustering, logistic regression, and random forest confirms biologically coherent risk patterns.

7. Fairness Evaluation

To assess potential demographic bias, AUC was evaluated by subgroup.

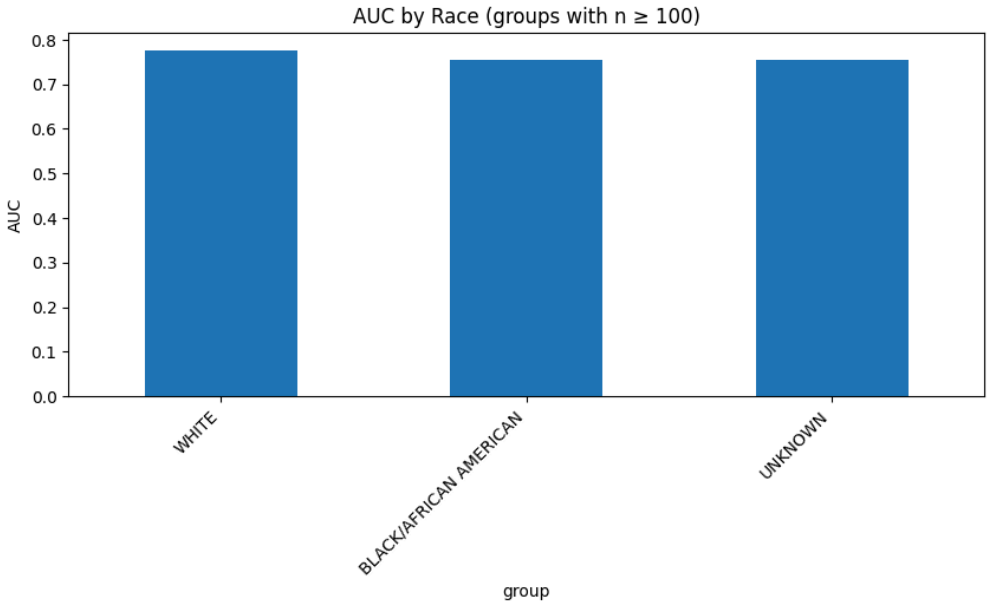
Gender:

- Male: 0.80
- Female: 0.77



Race (groups ≥100 samples):

- White: 0.776
- Black/African American: 0.756
- Unknown: 0.756



Differences were modest and may reflect sample size imbalance, documentation artifacts, or underlying healthcare disparities rather than model bias alone. These results highlight the importance of subgroup evaluation prior to clinical deployment.

8. Societal Impact and Ethical Considerations

Clinical Impact

Predictive models such as these could assist clinicians by identifying high-risk patients early in admission, enabling prioritized monitoring and intervention. However, overreliance on algorithmic outputs risks automation bias and reduced clinical autonomy.

Fairness and Representation

The dataset originates from a single academic medical center and does not represent the broader population. Underrepresentation of certain demographic groups may influence model behavior, emphasizing the need for continual evaluation and cautious interpretation.

Privacy and Re-identification Risk

Although Symile-MIMIC is de-identified, prior research demonstrates that anonymized healthcare data can sometimes be re-identified when combined with external sources. Unauthorized access could expose sensitive medical information. Credentialed access and governance frameworks are therefore essential.

Invisible Data Collection

Most patients are unaware that their laboratory tests and imaging studies may later be used for machine learning research. This highlights the growing role of passive data collection in healthcare and raises questions regarding transparency and informed consent.

Ethical Reflection

These models should function as decision-support tools rather than replacements for clinician judgment. Responsible deployment requires fairness auditing, privacy safeguards, and human oversight.

9. Limitations

Several limitations should be noted:

- Observational EHR data does not support causal claims.
- Missingness and measurement frequency vary by clinical severity.
- CXR timing (24–72 hours post-admission) complicates strict “early-only” prediction interpretation.
- Single-dataset setting; external validation across hospitals was not performed.
- Structured features used; deeper multimodal fusion (raw ECG/CXR encoders) was out of scope for DS-600.

Future work could incorporate longitudinal data, clinical notes, and external validation cohorts.

10. Conclusion

This project demonstrates that early laboratory measurements contain substantial signals for identifying high-risk patients and discovering clinically meaningful phenotypes. Unsupervised clustering revealed distinct severity profiles, while supervised models achieved moderate discrimination for in-hospital mortality.

Importantly, consistent physiologic drivers emerged across analytical methods, including metabolic acidosis, renal dysfunction, inflammation, and coagulopathy. While predictive models show promise, ethical deployment requires careful attention to fairness, privacy, and clinical integration.

Ultimately, data mining can enhance healthcare delivery when applied responsibly, transparently, and in partnership with human expertise.

References

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